Applying Science of Learning Activity

Theme 3 – Application of learning to new and varied contexts - Tip 5 - Tip Name: Explicitly prepare learners to transfer knowledge to new settings

Example 1:

This study describes the novel creation of a validation tool to assess learners skills using a simulation tool to aid in learning colonoscopy skills. Endoscopy requires a steep learning curve and is a difficult skill to even begin to practice. The use of simulation can aid with a learner's comfort with the use of a scope prior to transferring those skills to be used in humans. As described in tip 5, this study incorporates the need for advanced reasoning by learners which can be aided by educators in creating complex schemas through simulation.

Jaensch, C., Jensen, R. D., Paltved, C., & Madsen, A. H. (2023). Development and validation of a simulation-based assessment tool in colonoscopy. Advances in Simulation, 8(1). https://doi.org/10.1186/s41077-023-00260-5

Example 2:

The use of virtual reality in skill development in medicine and surgery is growing along with further research into this field. In this study, VR skills were assessed and compared amongst subjects with a history of gaming console use and those without. Outcomes were also prospectively followed following group assignment (asked to avoid gaming for 14 days vs those asked to game for 14 days) and skills retested. This is a valuable study to follow psychomotor skills and transference of these skills from console gaming to VR simulators which would be important in endoscopy training. By introducing these skills in a stepwise manner to prepare learners for endoscopy, this incorporates tip 5 through decreasing cognitive load through simulation/console gaming.

Radu Gugura, Fischer, P., Tanțău, M., & Cristian Tefas. (2023). Just five more minutes, mom: why video games could make you a better endoscopist. Surgical Endoscopy and Other Interventional Techniques, 37(9), 6901–6907. https://doi.org/10.1007/s00464-023-10167-x

Theme 5 – Harness the power of emotions for learning - Tip 8 – Teach learners to recognize their emotional state and its role in their learning

Example 1:

Practical medical education incorporates theoretical medical knowledge with real-life scenarios that often leave an emotional impact on learners, especially in critical care medicine. There are often emergent situations where one's decision-making in the moment can impact a patient's long-term outcome or survival. Schwartz rounds, as described in this study, allow healthcare professionals to reflect on the emotional toll of working in healthcare, and allow them to reflect, process, and learn from challenging cases. This study embodies tip #8 very well and allows learners to apply their knowledge to approach future cases differently (or to better prepare emotionally).

Ng, L., Schache, K., Young, M., & Sinclair, J. (2023). Value of Schwartz Rounds in promoting the emotional well-being of healthcare workers: a qualitative study. BMJ Open, 13(4), e064144. https://doi.org/10.1136/bmjopen-2022-064144

Example 2:

This literature review delves into the emotional challenges faced by medical trainees through medical school and residency, along with its impact on physician well-being and medical education. Both positive and negative emotional changes are investigated which is a strength to this study, and encompasses themes discussed in tip #8. The authors also elaborate on the emotions not only experienced in medical school, but throughout a learner's training that can impact their ability on knowledge transference throughout one's career. Educators being able to empathize with a learner's emotional state would aid in knowledge transference by making a topic more applicable/relatable to an individual based on their unique context as discussed in tip 8.

McConnell, M. M., & Eva, K. W. (2012). The Role of Emotion in the Learning and Transfer of Clinical Skills and Knowledge. Academic Medicine, 87(10), 1316–1322. https://doi.org/10.1097/ACM.0b013e3182675af2

Theme 6 – Teaching and learning in social contexts- Tip 10 – Attend to the social nature of learning

Example 1

This unique study involving Twitter investigates the incorporation of multimedia tweets (compared to 'standard' tweets) to promote readership through a randomized-controlled trial. The authors found that multimedia tweets lead to an increase in full-text downloads, as well as >1000 more impressions compared to standard tweets. Researchers should leverage the findings identified in this study, and to reach a global user/learner base understand the advantages present with multimedia tweets to promote their peer-reviewed articles that have been accepted for publication. This can also promote online public discussions regarding the research and interactions with the authors.

Tapper, E. B., Mirabella, R., Walicki, J., & Banales, J. (2021). Optimizing the use of twitter for research dissemination: The "Three Facts and a Story" Randomized-Controlled Trial. Journal of Hepatology. https://doi.org/10.1016/j.jhep.2021.05.020

Example 2

This article describes the impact of social media on medical education as demonstrated by altmetric scores and Mendeley downloads. Ultimately Mendeley downloads correlated the best with readership and citations within the Medical Education journal, the same impact was not seen with Twitter mentions. This is an important study for educators in particular as it is important to recognize not only how learners are exposed to new publications (i.e., the impact of social media), but it also relates to tip #10 through the discussion of potential bias where articles can be popular due to subject matter of the research (rather than due to specific outcomes/content of the paper), which can impact the impression that is made on learners.

Amath, A., Ambacher, K., Leddy, J. J., Wood, T. J., & Ramnanan, C. J. (2017). Comparing alternative and traditional dissemination metrics in medical education. Medical Education, 51(9), 935–941. https://doi.org/10.1111/medu.13359